

Rank Bounds and a Gauss-Fiber Alternative for Point-Span Tangent Absorption

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Abstract

Let $X^d \subset \mathbf{P}(V)$ be a smooth complex projective variety, let $H = \mathcal{O}_X(1)$, and embed X by the complete system H^m , where $m \geq 3$. A nonempty finite reduced set $Z \subset X$ is point-span tangent-absorbing if the vector span S_Z of its evaluation functionals contains the affine embedded tangent space at every point of Z . We prove

$$\dim_{\mathbf{C}} S_Z, |Z| \geq \min \left\{ (d+1)(d+2), \binom{d+m}{d} \right\}.$$

The proof gives two nonexclusive alternatives. Either Z contains $d+2$ projectively independent supports whose double points impose independent conditions, or Z lies in a common projective tangent d -plane and in one fiber of the original Gauss map. In the second branch, a characteristic-zero minimal-degree argument shows that the points are unisolvent in degree m on that plane. Consequently the stronger bound $(d+1)(d+2)$ is unconditional for $d \geq 3$, and also for $d = 2, m \geq 4$. For $d = 2, m = 3$, the exact minimum is 10, attained by a triangular lattice of ten points in a linearly embedded plane. Three points on a linearly embedded $\mathbf{P}^1$ for $m = 2$ show that the degree assumption cannot be weakened in the uniform theorem. A separate consequence of standard jet-ampleness gives a further lower bound that grows linearly with m and eventually improves the constant branch.

1 Introduction

Throughout, a variety is integral and all schemes and vector spaces are over $\mathbf{C}$. Let $X^d \subset \mathbf{P}(V)$ be smooth and let $H = \mathcal{O}_X(1)$. Write

$$\varphi_m = \varphi_{|H^m|} : X \longrightarrow \mathbf{P}(H^0(X, H^m)^*)$$

for the complete embedding defined by H^m , and put

$$S_Z := \text{Span}_{\mathbf{C}} \{ \text{ev}_p : p \in Z \} \subset H^0(X, H^m)^*.$$

We study nonempty finite reduced sets $Z \subset X$ satisfying

$$\widehat{T}_{\varphi_m(p)} \varphi_m(X) \subseteq S_Z \quad (p \in Z), \tag{1}$$

where the left side is the affine tangent space to the cone over the H^m -embedding. We use φ_m , rather than an ambient Veronese notation, because restriction of degree- m forms on $\mathbf{P}(V)$ need not be the complete linear series when X is not projectively normal. We call (1) *point-span tangent absorption*.

The condition is an extreme interpolation defect. The annihilator of the point span is the space of sections vanishing on Z , whereas the annihilator of the tangent span is the space of

sections vanishing on the union $2Z$ of the first infinitesimal neighborhoods. Thus absorption is equivalent to

$$H^0(X, \mathcal{I}_Z \otimes H^m) = H^0(X, \mathcal{I}_{2Z} \otimes H^m). \quad (2)$$

Write

$$\gamma_X : X \longrightarrow \mathrm{Gr}(d+1, V), \quad p \longmapsto \hat{T}_p X$$

for the Gauss map of the original embedding, and define

$$g_X(Z) := \max_{W \in \mathrm{Gr}(d+1, V)} |Z \cap \gamma_X^{-1}(W)|. \quad (3)$$

The main result is the following.

Theorem 1.1 (Rank bound and Gauss-fiber alternative). *Let $X^d \subset \mathbf{P}(V)$ be a smooth complex projective variety, $d \geq 1$, and let $m \geq 3$. If $Z \subset X$ is a nonempty finite reduced point-span tangent-absorbing set for the complete H^m -embedding, then Z contains $d+1$ supports whose first infinitesimal neighborhoods impose independent conditions on H^m . In particular,*

$$\dim S_Z \geq (d+1)^2, \quad |Z| \geq (d+1)^2. \quad (4)$$

Moreover, at least one of the following nonexclusive alternatives holds:

- (a) Z contains $d+2$ supports whose first infinitesimal neighborhoods impose independent conditions on H^m ; hence

$$\dim S_Z, |Z| \geq (d+1)(d+2);$$

- (b) there is a vector subspace $W \subset V$, $\dim W = d+1$, such that

$$Z \subset X \cap \mathbf{P}(W), \quad \hat{T}_p X = W \quad (p \in Z),$$

and in this case

$$\dim S_Z, |Z| \geq \binom{d+m}{d}.$$

Consequently,

$$\boxed{\dim S_Z, |Z| \geq \min \left\{ (d+1)(d+2), \binom{d+m}{d} \right\}.} \quad (5)$$

The two numerical terms come from different mechanisms. The first is a finite first-jet independence statement. We derive the relevant statement from Ballico–Chiantini’s criterion [2, Theorem 3.5] and retain a direct squared-hyperplane proof of this specialization. The second term is the dimension of degree- m forms on the common tangent $\mathbf{P}^d$ in the Gauss branch. Its proof uses characteristic zero essentially.

Corollary 1.2 (Numerical consequences). *Under the hypotheses of Theorem 1.1:*

- (i) if $d \geq 3$, then $\dim S_Z, |Z| \geq (d+1)(d+2)$ for every $m \geq 3$;
- (ii) if $d = 2$ and $m \geq 4$, then $\dim S_Z, |Z| \geq 12$;
- (iii) if $d = 2$ and $m = 3$, then $\dim S_Z, |Z| \geq 10$, and this is exact;
- (iv) for every $d \geq 2$ equality in the baseline $\dim S_Z = (d+1)^2$ is impossible.

Corollary 1.3 (Gauss-fiber threshold). *If*

$$g_X(Z) < \binom{d+m}{d},$$

then alternative (a) of Theorem 1.1 holds. In particular, $\dim S_Z, |Z| \geq (d+1)(d+2)$.

Corollary 1.4 (Jet-ampleness supplement). *Under the hypotheses of Theorem 1.1, set*

$$J(d, m) := \left\lfloor \frac{m+1}{2} \right\rfloor (d+1) + \begin{cases} 1, & m \text{ even,} \\ 0, & m \text{ odd.} \end{cases}$$

Then

$$\dim S_Z, |Z| \geq \max \left\{ J(d, m), \min \left((d+1)(d+2), \binom{d+m}{d} \right) \right\}.$$

In particular, $J(d, m) > (d+1)(d+2)$ for every $m \geq 2d+4$. This is a lower bound, not an assertion of exactness.

The paper makes no claim that the framework of first-jet independence is new. The contribution asserted here is limited to the stated deduction for point-span absorption and the elementary estimate in its exceptional Gauss branch. The bibliographic comparison in Section 9 is selective rather than a certification of priority.

2 First jets and absorption

For a smooth point $p \in X$, let $2p$ be the first infinitesimal neighborhood defined by $\mathfrak{m}_p^2$. It has length $d+1$. Since H^m is very ample, restriction to $2p$ is surjective. Intrinsically, its dual is the affine tangent space to the cone over the H^m -embedding; this description avoids any coordinate-dependent scalar in a symmetric-tensor formula.

Definition 2.1. *A nonempty finite reduced set $Z \subset X$ is point-span tangent-absorbing with respect to H^m if $\hat{T}_{\varphi_m(p)}\varphi_m(X) \subseteq S_Z$ for every $p \in Z$.*

Lemma 2.2 (Kernel criterion). *The following are equivalent:*

- (i) *Z is point-span tangent-absorbing;*
- (ii) *equation (2) holds;*
- (iii) *restriction to Z and restriction to $2Z$ have the same image rank.*

Proof. The annihilator of S_Z is the kernel of restriction to Z . The annihilator of the span of the affine tangent spaces is the kernel of restriction to $2Z$. Since the point span is contained in the tangent span, the three conditions are equivalent. $\square$

3 Independent supports and first-jet blocks

We first isolate the precise independence input. This is a specialization of a published general criterion, not a new general theorem about jets.

Lemma 3.1 (Projectively independent supports). *Let $S \subset X$ be a finite set whose representatives in V are linearly independent. If $m \geq 3$, then $2S$ imposes independent conditions on H^m .*

Proof. We first explain the specialization of Ballico–Chiantini [2, Definition 3.1 and Theorem 3.5]. Their condition $\dagger$ for (X, L_1, S) says that, for every $p \in S$, the restriction map

$$H^0(X, L_1) \longrightarrow H^0(S \cup 2p, L_1|_{S \cup 2p})$$

is surjective. Take

$$L_1 = H^2, \quad L_2 = H^{m-2}, \quad L_1 \otimes L_2 = H^m.$$

For each $q \in S$, projective independence gives a linear form h_q that vanishes on $S \setminus \{q\}$ and not at q . The sections h_q^{m-2} show that S imposes independent conditions on L_2 .

Fix $p \in S$. For $q \neq p$, the square h_q^2 prescribes the value at q and vanishes on $2p$. The square h_p^2 supplies the value at p . If $a \in V^*$ vanishes at p , then $h_p a$ vanishes at every support and has differential $h_p(p) da$ at p . Because H embeds X , the restrictions of such differentials span the cotangent space of X at p . These quadrics therefore give surjectivity on $S \cup 2p$, so (X, H^2, S) satisfies $\dagger$. Ballico–Chiantini’s Theorem 3.5 now implies that $2S$ imposes independent conditions on H^m .

For completeness, the same specialization has the following direct squared-hyperplane proof. Order $S = \{p_1, \dots, p_r\}$. A single double point is independent because H^m is very ample. Suppose the doubles at $p_1, \dots, p_{j-1}$ are independent. Choose a linear form e vanishing on their representative span and nonzero at p_j . Then e^2 vanishes on all old double points and is a unit on $2p_j$. Since H^{m-2} is very ample, multiplication by e^2 maps a space of sections surjectively onto the first jets at p_j . It can therefore correct arbitrary residual data at $2p_j$ without changing the old data. Induction proves the assertion. This direct proof is included to keep the exact specialization self-contained. $\square$

This lemma concerns the displayed finite set of supports and retains a direct proof tailored to that configuration. The separate global fact that H^m is m -jet ample yields Corollary 1.4; its source and role are made explicit in the proof of that corollary.

4 Escape or Gauss-fiber concentration

The next separator is formulated with first jets. It uses only ambient linear forms and remains valid without projective normality.

Lemma 4.1 (Tangent escape separator). *Let $0 \neq v \in W \subsetneq V$ represent $p = [v] \in X$, and let $u \in \widehat{T}_p X \setminus W$. For every $m \geq 1$ there is a section of H^m that vanishes at every point of $X \cap \mathbf{P}(W)$ but has nonzero first jet at p in the direction u .*

Proof. Choose $\lambda \in V^*$ with $\lambda|_W = 0$ and $\lambda(u) \neq 0$, and choose $\mu \in V^*$ with $\mu(v) \neq 0$. The restriction to X of $f = \lambda\mu^{m-1}$ vanishes on $X \cap \mathbf{P}(W)$, while

$$df_v(u) = \lambda(u)\mu(v)^{m-1} \neq 0.$$

Thus its first jet separates the indicated tangent direction from every linear combination of evaluations supported on $X \cap \mathbf{P}(W)$. $\square$

Lemma 4.2 (Support alternative). *Assume Z is point-span tangent-absorbing. Let $p_1, \dots, p_r \in Z$ have linearly independent representatives and put $W = \text{Span}(v_1, \dots, v_r)$.*

- (i) *If $r \leq d$, some point of Z lies outside $\mathbf{P}(W)$.*
- (ii) *If $r = d + 1$, either some point of Z lies outside $\mathbf{P}(W)$, or*

$$Z \subset \mathbf{P}(W), \quad \widehat{T}_p X = W \quad (p \in Z).$$

Proof. Suppose $Z \subset \mathbf{P}(W)$. If $r \leq d$, then $\dim W = r < d + 1 = \dim \widehat{T}_{p_1} X$, so there is $u \in \widehat{T}_{p_1} X \setminus W$. Lemma 4.1 gives a section annihilating S_Z but not the corresponding tangent functional, contrary to absorption.

Now let $r = d + 1$. Both W and every $\widehat{T}_p X$ have dimension $d + 1$. If $\widehat{T}_p X \neq W$ for some $p \in Z$, choose $u \in \widehat{T}_p X \setminus W$ and apply the same separator. Hence the no-escape case forces $\widehat{T}_p X = W$ for every $p \in Z$. $\square$

5 Unisolvence in the Gauss branch

The following lemma is the additional characteristic-zero input.

Lemma 5.1 (Gauss-branch unisolvence). *Let W have dimension $d + 1$, let $Z \subset \mathbf{P}(W)$ be a nonempty finite reduced set, and suppose*

$$\widehat{T}_p X = W \quad (p \in Z).$$

If Z is point-span tangent-absorbing for H^m , then, for $R = \text{Sym}(W^)$ and the homogeneous ideal $I = I(Z) \subset R$,*

$$I_m = 0.$$

Consequently,

$$\dim S_Z \geq \binom{d+m}{d}, \quad |Z| \geq \binom{d+m}{d}.$$

Proof. Let $F \in I_m$. The inclusion $W \subseteq V$ induces a surjection $V^* \rightarrow W^*$ and hence a surjection $\text{Sym}^m(V^*) \rightarrow \text{Sym}^m(W^*)$, so F extends to a degree- m form on V . Its restriction to X vanishes on Z . Under the natural pairing, the resulting section pairs to zero with every evaluation in S_Z . Absorption implies that its first jet vanishes along X at every $p \in Z$. Because $\widehat{T}_p X = W$, this is the full first jet of F on $\mathbf{P}(W)$.

To identify that condition scheme-theoretically, choose homogeneous coordinates $x_0, \dots, x_d$ on W with $p = [1 : 0 : \dots : 0]$. On the chart $x_0 = 1$, vanishing of the value and of all projective directional derivatives says that $F(1, x_1, \dots, x_d)$ has no terms of degree zero or one. It therefore lies in $(x_1, \dots, x_d)^2$; by homogeneity, this is equivalent to $F \in I(p)^2$. Consequently

$$I_m \subseteq \bigcap_{p \in Z} I(p)^2. \quad (6)$$

Suppose $I_m \neq 0$. Choose a nonzero homogeneous $F \in I$ of minimal positive degree α . Then $\alpha \leq m$. For each $p \in Z$, choose $G_p \in R_{m-\alpha}$ with $G_p(p) \neq 0$; when $\alpha = m$, take a nonzero constant. Since $G_p F \in I_m$, equation (6) gives $G_p F \in I(p)^2$. The germ of G_p is a unit at p , hence $F \in I(p)^2$ in the local ring at p .

It follows that every first partial derivative of F vanishes at every point of Z . Thus each partial derivative belongs to the radical ideal $I(Z)$. These derivatives have degree $\alpha - 1$, and in characteristic zero at least one is nonzero because F has positive degree. This contradicts the minimality of α . Hence $I_m = 0$.

The evaluation map

$$R_m \longrightarrow \mathbf{C}^Z$$

is therefore injective and has rank $\dim R_m = \binom{d+m}{d}$. Every form on W extends to V , so this value matrix is a submatrix of the restriction map for H^m . Its rank is at most both $\dim S_Z$ and $|Z|$, proving the inequalities. $\square$

Remark 5.2. *Characteristic zero is used at the exact point where a positive-degree polynomial is required to have a nonzero partial derivative. In characteristic p , a p th power defeats that step. The separator and the squared-hyperplane construction are scheme-theoretic and appear characteristic-free, but Theorem 1.1 is not asserted here in positive characteristic.*

6 Proof of the main theorem

Proof of Theorem 1.1. Start with any point of Z . Repeated application of Lemma 4.2(i) selects $d + 1$ points with linearly independent representatives. Lemma 3.1 shows that their double points impose independent conditions on H^m .

Dualizing the restriction map to these double points embeds a direct sum of $d + 1$ vector spaces of dimension $d + 1$ in $H^0(X, H^m)^*$. Each summand is an affine tangent space and hence lies in S_Z by absorption. Therefore

$$\dim S_Z \geq (d + 1)^2.$$

Since S_Z is spanned by $|Z|$ evaluation vectors, also $|Z| \geq \dim S_Z$. This proves (4).

Apply Lemma 4.2(ii) to the selected $d + 1$ points. If another point escapes their representative span, the resulting $d + 2$ supports are projectively independent. Lemma 3.1 then gives alternative (a) and the bound $(d + 1)(d + 2)$.

If no point escapes, Lemma 4.2(ii) gives the subspace W and the common Gauss fiber in alternative (b). Apply Lemma 5.1 to obtain the bound $\binom{d+m}{d}$. Since at least one branch holds, taking the smaller of their two bounds proves (5). $\square$

Proof of Corollary 1.2. For $d \geq 3$ and $m \geq 3$,

$$\binom{d+m}{d} \geq \binom{d+3}{d} = \frac{(d+1)(d+2)(d+3)}{6} \geq (d+1)(d+2).$$

For $d = 2$ and $m \geq 4$, one has $\binom{m+2}{2} \geq 15 > 12$. For $(d, m) = (2, 3)$ the two terms in (5) are 12 and 10; exactness is proved in Proposition 7.1 below.

Finally, $(d + 1)(d + 2) > (d + 1)^2$, while for $d \geq 2$

$$\binom{d+m}{d} \geq \binom{d+3}{d} > (d+1)^2,$$

the last strict inequality being equivalent to $d(d - 1) > 0$. Thus equality in the square baseline is impossible for $d \geq 2$. $\square$

Proof of Corollary 1.3. In alternative (b), all of Z lies in a single Gauss fiber, and Lemma 5.1 gives $g_X(Z) = |Z| \geq \binom{d+m}{d}$. The displayed strict hypothesis therefore excludes alternative (b), so alternative (a) must hold. $\square$

Proof of Corollary 1.4. A very ample line bundle is 1-jet ample, and Beltrametti–Di Rocco–Sommese [3, Proposition 2.3] prove that the tensor product of an a -jet ample bundle and a b -jet ample bundle is $(a + b)$ -jet ample. Thus H^m is m -jet ample. In their convention, a simple point has weight one, a first neighborhood $2p$ has weight two, and surjectivity is required for every collection of positive weights summing to $m + 1$.

Suppose first that $m = 2q + 1$. If $|Z| \leq q$, enlarge $2Z$ at one support to a fat-point scheme of total weight $m + 1$. Surjectivity onto that larger scheme implies surjectivity onto $2Z$. The dual tangent blocks would then be independent of total dimension $|Z|(d + 1)$, but absorption places their span inside S_Z , whose dimension is at most $|Z|$. This is impossible because $d \geq 1$ and Z is nonempty. Hence Z contains $q + 1$ distinct supports. Their first neighborhoods have total weight $2(q + 1) = m + 1$, so their tangent blocks are independent and contained in S_Z . Therefore

$$\dim S_Z, |Z| \geq (q + 1)(d + 1) = J(d, m).$$

If $m = 2q$, the same thickening argument excludes $|Z| \leq q$. Choose q supports and one further support. The union of the first neighborhoods of the first q points and the simple

last point has total weight $2q + 1 = m + 1$. Jet ampleness makes the corresponding dual blocks independent; they lie in S_Z by absorption and by the definition of the point span. Consequently

$$\dim S_Z, |Z| \geq q(d + 1) + 1 = J(d, m).$$

Combining this estimate with Theorem 1.1 gives the displayed maximum. Finally, direct substitution in the even and odd cases shows that $J(d, m) > (d + 1)(d + 2)$ for every $m \geq 2d + 4$. $\square$

7 Exact examples and the degree threshold

Proposition 7.1 (Exact plane-cubic minimum). *For a linearly embedded $X = \mathbf{P}^2$ and $m = 3$, the minimum cardinality and minimum vector-span dimension of a nonempty reduced point-span tangent-absorbing set are both 10. Equality is attained by*

$$Z = \{[1 : i : j] : i, j \in \mathbf{Z}_{\geq 0}, i + j \leq 3\}.$$

Proof. The lower bound 10 is Corollary 1.2(iii). The displayed set has ten points. To prove that its evaluations span $H^0(\mathbf{P}^2, \mathcal{O}(3))^*$, dehomogenize and let $f(x, y)$ have total degree at most three and vanish on the triangular lattice.

On the layer $y = 0$, the polynomial $f(x, 0)$ has the four roots $0, 1, 2, 3$, hence vanishes identically and $f = yf_1$ with $\deg f_1 \leq 2$. On $y = 1$, the polynomial $f_1(x, 1)$ has the three roots $0, 1, 2$, so $f_1 = (y - 1)f_2$ with $\deg f_2 \leq 1$. On $y = 2$, $f_2(x, 2)$ has the two roots $0, 1$, so $f_2 = (y - 2)f_3$ with f_3 constant. The last point on $y = 3$ forces $f_3 = 0$. Thus $f = 0$.

The evaluation map between two ten-dimensional vector spaces is an isomorphism. Hence the point span is the full ambient dual and contains every tangent space. Therefore Z is absorbing and $\dim S_Z = |Z| = 10$. $\square$

Lemma 7.2 (Simplex-lattice unisolvence). *For integers $n, r \geq 0$, the set*

$$\Delta_{n,r} := \{(a_1, \dots, a_n) \in \mathbf{Z}_{\geq 0}^n : a_1 + \dots + a_n \leq r\}$$

is unisolvent for polynomials in n variables of total degree at most r .

Proof. We induct on $n + r$. The cases $n = 0$ and $r = 0$ are immediate. Suppose that f of degree at most r vanishes on $\Delta_{n,r}$. Its restriction $f(x_1, \dots, x_{n-1}, 0)$ vanishes on $\Delta_{n-1,r}$ and is zero by induction. Hence $f = x_n g$ with $\deg g \leq r - 1$. For every $(a_1, \dots, a_{n-1}, s) \in \Delta_{n,r}$ with $s \geq 1$, one has $g = 0$ at that point. Thus

$$h(x_1, \dots, x_{n-1}, t) := g(x_1, \dots, x_{n-1}, t + 1)$$

vanishes on $\Delta_{n,r-1}$. Induction gives $h = 0$, hence $g = 0$ and $f = 0$. Since $|\Delta_{n,r}| = \binom{n+r}{n}$ equals the dimension of the polynomial space, the evaluation map is an isomorphism. $\square$

Remark 7.3 (A three-dimensional equality fixture). *For $X = \mathbf{P}^3$ and $m = 3$, the bound is 20 and is also attained by*

$$\{[1 : i : j : k] : i, j, k \in \mathbf{Z}_{\geq 0}, i + j + k \leq 3\}.$$

Lemma 7.2, with $(n, r) = (3, 3)$, proves that these twenty evaluations are unisolvent for polynomials of total degree at most three. Their span is ambient, so the fixture is absorbing.

Proposition 7.4 (Degree two cannot satisfy the uniform baseline). *Let $X = \mathbf{P}^1$ in its linear embedding and let $m = 2$. Any three distinct points form a point-span tangent-absorbing set with*

$$\dim S_Z = |Z| = 3 < 4 = (d+1)^2.$$

Consequently the hypothesis $m \geq 3$ in the uniform baseline cannot be weakened to $m = 2$.

Proof. A binary quadratic vanishing at three distinct points is zero. Hence evaluation at the three points is an isomorphism

$$H^0(\mathbf{P}^1, \mathcal{O}(2)) \longrightarrow \mathbf{C}^3.$$

The three evaluation vectors span the full ambient dual, so they contain all embedded tangent spaces and absorption is automatic. The asserted dimensions follow. $\square$

For comparison, when $X = \mathbf{P}^1$ and $m = 3$, four distinct points fill $H^0(\mathbf{P}^1, \mathcal{O}(3))^*$ and attain the square baseline 4. Thus the square bound remains exact in dimension one, although Corollary 1.2 shows that it is never exact in dimensions at least two.

8 Smooth quadrics

Corollary 8.1. *Let $Q^d \subset \mathbf{P}^{d+1}$ be a smooth complex quadric and let $m \geq 3$. Every nonempty finite reduced point-span tangent-absorbing set for $\mathcal{O}_Q(m)$ satisfies*

$$\dim S_Z, |Z| \geq (d+1)(d+2).$$

Proof. If B is a nondegenerate symmetric bilinear form defining Q , then the affine tangent space at $[v]$ is the polar hyperplane $v^\perp$. Polarity is injective, so every Gauss fiber has one point. Since $1 < \binom{d+m}{d}$, Corollary 1.3 applies. $\square$

For a quadric curve $Q^1 \cong \mathbf{P}^1$ one has $\mathcal{O}_Q(m) \cong \mathcal{O}_{\mathbf{P}^1}(2m)$. A reduced set is absorbing exactly when it has at least $2m+1$ points: at that cardinality its evaluations fill the ambient dual, while with at most $2m$ points one can construct a degree- $2m$ form vanishing simply at any prescribed support. This sharper one-dimensional calculation is compatible with, but not implied sharply by, the dimension-independent quadric bound above.

9 Relation to adjacent interpolation notions

Terracini loci parameterize finite sets for which the union of double points fails to impose the expected independent conditions [2, 6]. Point-span absorption is more restrictive: the value rank and the double-point rank are equal. It therefore produces an extreme Terracini defect whenever the value span is not already ambient, but ordinary Terracini defect does not imply absorption.

The independence lemma belongs to the established theory of simultaneous jet separation. In the terminology of Beltrametti–Di Rocco–Sommese [3], jet ampleness asks for uniform surjectivity over prescribed zero-dimensional data. Their Proposition 2.3 implies that H^m is m -jet ample and is exactly the external input used in Corollary 1.4. By contrast, the proof of Lemma 3.1 separates only its specific finite union of first neighborhoods. Ballico–Chiantini’s condition $\dagger$ and their Theorem 3.5 provide exactly the product criterion used in Lemma 3.1.

Base loci and strong base loci of tangential projections concern spans generated by tangent spaces that contain an additional point or an additional tangent space [1]. There is an exact

translation in one direction. Since the evaluation at p lies in the affine tangent space at $\varphi_m(p)$, absorption gives

$$S_Z = \text{Span}\{\widehat{T}_{\varphi_m(p)}\varphi_m(X) : p \in Z\}.$$

If $A \subseteq Z$ is minimal among subsets whose tangent spaces span S_Z , then every $q \in Z \setminus A$ belongs to the strong base locus of A for $\varphi_m(X)$. The extra feature here is that this tangent span equals the span of evaluations of the full support Z . The Veronese classifications in [1] therefore give relevant context, but do not by themselves establish the point-span bounds proved above. That source also shows that strong base loci and Terracini loci are not comparable in general.

Finally, the exceptional branch uses the Gauss map of the original embedding, not the Gauss map after the m th re-embedding. General fibers of Gauss maps in characteristic zero and the additional inseparable phenomena in positive characteristic belong to a broader classical theory; see, for example, [4, 5]. No classification of Gauss-degenerate varieties or their absorbing configurations is attempted here.

These comparisons delimit the claims. They do not establish that no related statement exists elsewhere, and no exhaustive priority assertion is made.

10 Scope and limitations

- Theorem 1.1 is stated and proved over $\mathbf{C}$. The minimal-degree derivative step in Lemma 5.1 is the specific obstruction to carrying this proof unchanged to positive characteristic.
- The theorem bounds rank and cardinality but does not classify absorbing sets, even inside the common Gauss fiber of alternative (b).
- The plane and three-space equality examples fill their ambient value spaces. No assertion is made that all equality cases are ambient or that the examples classify equality.
- The comparison with the cited literature is deliberately conservative. It supplies attribution and logical distinctions, not a comprehensive bibliographic or priority review.
- Finite computational rank checks can verify explicit fixtures but do not prove the universal theorem. The proof is the argument in Sections 3–6.

11 Conclusion

Point-span tangent absorption forces two complementary forms of rigidity. Support escape produces $d + 2$ independent first-jet blocks, while failure of escape concentrates the set in a common tangent plane and forces degree- m unisolvence there. Combining the two branches yields the minimum in (5), the unconditional strong range in Corollary 1.2, and the exact plane-cubic value 10. Jet-ampleness supplies the complementary growing estimate $J(d, m)$ of Corollary 1.4. The argument also isolates why characteristic zero and the threshold $m \geq 3$ enter the present formulation.

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